Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P09

Set 9

Time: 25 minutes | Paper Reference **PMAT2/N09**

Mathematics

Level 2
Section A (Non–Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

P69192A

©2021 Pearson Education Ltd.

1/1/1

P 6 9 1 9 2 A 0 1 0 8

Pearson

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Shannan has a new car.

The petrol tank has a maximum capacity of 48 litres.
The cost of petrol is 126.7p per litre.

Shannan thinks it will cost less than £70 to completely fill an empty tank with petrol.

Use **estimation** to check if Shannan is correct.
You **must** show your working.

(3)

Tutor Golden Rule: Estimation means rounding to 1 or 2 sig figs BEFORE calculating!

Step 1: Round values to convenient numbers (1 or 2 sig figs)

- **Fuel capacity: 48 litres ≈ 50 litres** *(rounded to nearest 10)*
- **Petrol price: 126.7p ≈ 130p = £1.30** *(or 1.30 per litre)*

Step 2: Calculate estimated total cost to fill tank

Estimated Cost = 50 litres × £1.30

Working:

50 × £1.00 = £50.00
50 × £0.30 = £15.00

£50.00 + £15.00 = £65.00 [M1]

Step 3: Compare with Shannan's estimate (£70)

£65.00 < £70.00 [M1 A1]

Since £65 is less than £70, Shannan is correct!

Yes, Shannan is correct

[3 Marks: M1 M1 A1]

(Total for Question 1 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2 Henry is designing the car park for a new shopping centre.
The total number of parking spaces at the shopping centre will be 900

150 of the parking spaces will be for staff.
The rest of the parking spaces will be for visitors.

Henry wants 6% of the visitor parking spaces to be parking spaces for disabled visitors.

Work out the number of parking spaces for disabled visitors in this car park.

(3)

Step 1: Find the number of visitor parking spaces

Total spaces = 900

Staff spaces = 150

Visitor spaces = 900 − 150 = 750 visitor spaces [M1]

Step 2: Calculate 6% of the 750 visitor spaces

Mental Chunking Method (Non-Calculator):

10% of 750 = 750 ÷ 10 = 75

1% of 750 = 75 ÷ 10 = 7.5

6% of 750 = 6 × 7.5 = (6 × 7) + (6 × 0.5)

= 42 + 3 = 45 spaces [M1]

Alternative Fraction Method:

6% × 750 = (6 ÷ 100) × 750 = 4,500 ÷ 100 = 45

45

[3 Marks: M1 M1 A1]

(Total for Question 2 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3 Here is a list of numbers.

12.96 12.096 12.9 12.45 12.1 12.738 12.62

(a) Find the median.

Tutor Golden Rule: To find the median, order numbers from smallest to largest!

(2)

Pad decimals with trailing zeros to 3 decimal places for easy comparison:

~~12.096~~, ~~12.100~~, ~~12.450~~, 12.620, ~~12.738~~, ~~12.900~~, ~~12.960~~

Middle value (4th of 7) = 12.62 [M1 A1]

12.62

(b) Work out $\frac{7}{8} - \frac{3}{5}$

You **must** show your working.

(2)

Step 1: Find the lowest common denominator: LCM(8, 5) = 40 [M1]

$\frac{7}{8} \times 5 = \frac{35}{40}$ $\quad$ $\frac{3}{5} \times 8 = \frac{24}{40}$

Step 2: Subtract the numerators over the common denominator

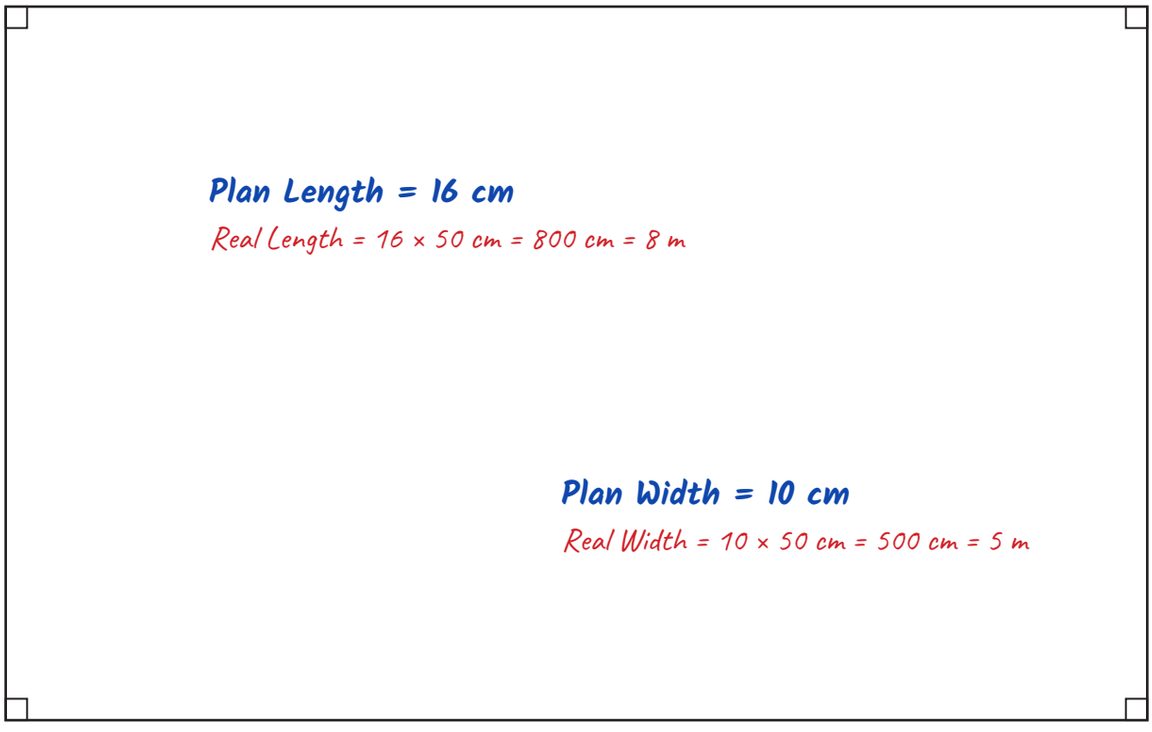
$\frac{35}{40} - \frac{24}{40} = \frac{35 - 24}{40} = \frac{11}{40}$ *[M1 A1] (11 is prime; fully simplified)*

$\frac{11}{40}$

(Total for Question 3 is 4 marks)

4 Raffi wants a two-storey extension built on the side of his house.

This scale diagram shows the plan view of the floor for each storey.



Scale 1 : 50

Raffi looks at the websites for different builders.

The table shows the cost per m^2 to build a one-storey extension.

Builder	A	B	C	D	E	F	G	H
Cost per m^2 (£)	1300	1350	1300	1325	1300	1340	1350	1290

Raffi will use the mode of these costs to help work out the expected cost to build this two-storey extension.

He also knows that the cost per m^2 to build a two-storey extension is $1\frac{1}{2}$ times the cost per m^2 to build a one-storey extension.

Cost frequencies: £1290 (1), £1300 (3), £1325 (1), £1340 (1), £1350 (2)

MODE = £1,300 per m^2 (most frequent builder cost, appears 3 times: A, C, E)

Turn over to Page 7 for full step-by-step cost calculations ->

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 9 1 9 2 A 0 6 0 8

Work out the expected cost to build the two-storey extension for Raffi.
You **must** show your working.

(6)

Step 1: Convert plan dimensions to real metres (Scale 1 : 50)

1 cm on plan represents 50 cm = 0.5 m in real life

- **Real Length = 16 cm × 50 = 800 cm = 8 m [M1]**
- **Real Width = 10 cm × 50 = 500 cm = 5 m [M1]**

Step 2: Calculate floor area for one storey

Floor Area = Length × Width = 8 m × 5 m = 40 m² [M1]

Step 3: Identify the mode cost per m² from table

Mode cost = £1,300 per m² (most frequent value) [M1]

Step 4: Calculate cost per m² for a two-storey extension

Cost per m² (2-storey) = 1.5 × £1,300

Working: 1 × £1,300 = £1,300

0.5 × £1,300 = £650

£1,300 + £650 = £1,950 per m² [M1]

Step 5: Calculate total expected cost for the extension

Total Expected Cost = Floor Area × Cost per m² (2-storey)

= 40 m² × £1,950 [M1]

Working: 4 × 1,950 = 7,800 -> 40 × 1,950 = £78,000 [A1]

Check via 1-storey area: 40 × £1,300 = £52,000; £52,000 × 1.5 = £78,000

£ **78,000**

[6 Marks: M1 M1 M1 M1 M1 A1]

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P09

Set 9

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C09**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 48 /48

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

©2021 Pearson Education Ltd.

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Thalia owns a chocolate factory.

The workers in the factory work 7 hours per day.
They produce 65 boxes of chocolate per hour.

Thalia gets an order for 9100 boxes of chocolates.

Work out the number of days needed to produce 9100 boxes of chocolates.

(3)

Method 1: Find daily factory output first (Recommended)

Step 1: Calculate total boxes produced per day

Boxes per day = 7 hours/day × 65 boxes/hour [M1]

Working: 7 × 60 = 420 , 7 × 5 = 35

420 + 35 = 455 boxes per day

Step 2: Calculate number of days needed for 9,100 boxes

Days needed = Total order ÷ Daily output

= 9,100 ÷ 455 [M1]

Notice: 910 ÷ 455 = 2 , so 9,100 ÷ 455 = 20 days [A1]

Method 2 (Alternative: Total hours required):

- Total hours = 9,100 ÷ 65 = 140 hours
- Days needed = 140 ÷ 7 = 20 days

Tutor Check: 20 days × 7 hrs/day × 65 boxes/hr = 9,100 boxes (correct)

20

[3 Marks: M1 M1 A1]

(Total for Question 1 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2 Thalia has 6 machines to produce chocolate in her factory.
Each machine produces the same amount of chocolate.

It takes $3\frac{1}{2}$ hours for these 6 machines to produce enough chocolate to fill a crate with chocolate bars.

On Tuesday only 4 of the machines are working.

Work out the number of hours the 4 machines will take to produce the chocolate needed to fill a crate with chocolate bars.

(3)

Tutor Golden Rule: INVERSE PROPORTION!

Fewer machines produce at a slower rate, so it takes MORE time to fill a crate.
The total amount of machine-hours required remains CONSTANT.

Step 1: Find total machine-hours required to produce one crate

Total work = 6 machines × 3.5 hours

Working: 6 × 3 = 18 , 6 × 0.5 = 3

18 + 3 = 21 machine-hours [M1]

Step 2: Calculate time needed when only 4 machines are running

Hours required = Total machine-hours ÷ 4 machines

= 21 ÷ 4 [M1]

= 5.25 hours (or 5 1/4 hours) [A1]

Converting decimal fraction of an hour to minutes:

0.25 hours = 0.25 × 60 minutes = 15 minutes

So 5.25 hours = 5 hours 15 minutes

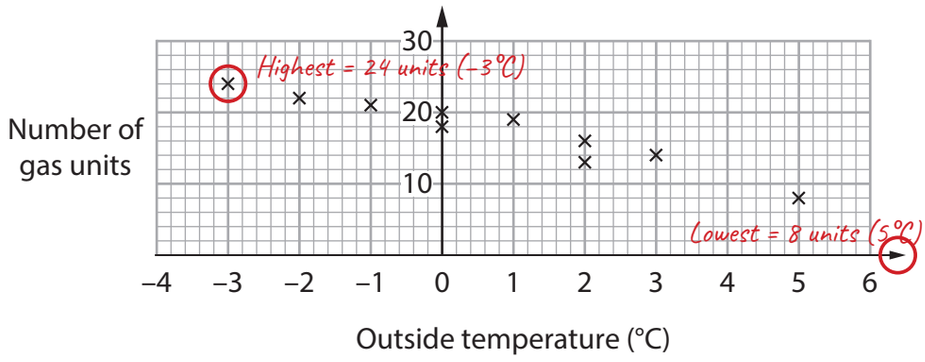
5.25

[3 Marks: M1 M1 A1]

(Total for Question 2 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3 The scatter diagram shows information about the outside temperature and the number of gas units used to heat a house on each of ten days during January.



The range of the number of gas units used to heat this house each day during October was 11 gas units per day.

Jenson thinks the number of gas units used to heat this house each day is more consistent in October than in the ten days in January.

(a) Is Jenson correct?
Show why you think this.

(3)

January: Highest = 24, Lowest = 8 -> Range = 24 − 8 = 16 gas units [M1]

October Range = 11 | January Range = 16 [M1]

October has smaller range (11 < 16), so October usage was more consistent. [A1]

Therefore, Jenson is correct.

Yes *[M1 M1 A1]*

Jenson also thinks there is no relationship between the outside temperature and the number of gas units used to heat this house in these ten days in January.

(b) Is Jenson correct?
Give a reason for your answer.

(1)

No, Jenson is incorrect. [A1]

There is a clear relationship: a negative correlation.

As the outside temperature increases, the number of gas units used decreases.

(Points follow a downward trend from left to right)

[4 Marks Total]

(Total for Question 3 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4 Kady runs a household waste centre.
Rubbish collected at the centre is either sent for recycling or sent to landfill.

The table shows information about the weight of each type of rubbish sent for recycling last month.

Rubbish type	Weight (tonnes)
Garden waste	224.56
Wood and cardboard	101.81
Metal	37
Glass, soil and rubble	138.9
Electrical	16.54
Other	82.45

The total weight of rubbish collected at the centre last month was 653.64 tonnes.

What percentage of all the rubbish collected last month was sent for recycling?
Give your answer correct to the nearest whole number.

(3)

Step 1: Calculate total weight of rubbish sent for recycling

Sum of all recycled types:

224.56 + 101.81 + 37.00 + 138.90 + 16.54 + 82.45

Total Recycled = 601.26 tonnes [M1]

Step 2: Express recycled weight as a percentage of total collected

Percentage = (Total Recycled ÷ Total Collected) × 100

= (601.26 ÷ 653.64) × 100 [M1]

= 0.919864... × 100 = 91.986...%

Step 3: Round correct to the nearest whole number

The first decimal digit is 9 (≥ 5), so round UP:

91.986...% ≈ 92% [A1]

92 %

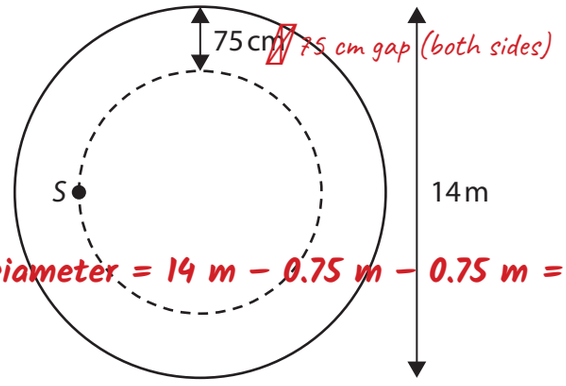
[3 Marks: M1 M1 A1]

(Total for Question 4 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5 Natalie will plant flower bulbs on a road roundabout.
The roundabout is in the shape of a circle with diameter 14 m.

Natalie will plant the flower bulbs in a circle 75 cm from the edge of the roundabout.
The dashed line in the sketch below shows where the flower bulbs will be planted.



Inner Diameter = 14 m − 0.75 m − 0.75 m = 12.5 m

Natalie will plant a flower bulb every 25 cm starting from point *S*.

The flower bulbs are sold in packs.
Each pack has 16 flower bulbs.

How many packs of flower bulbs will Natalie need?

(5)

Key Roadmap for Solution (Turn to Page 15 for full working):

1. Convert gap: 75 cm = 0.75 m. Subtract 0.75 m from BOTH sides.
2. Inner planting diameter = 14 − (2 × 0.75) = 12.5 m (or 1,250 cm).
3. Circumference = π × d = π × 12.5 m = 39.27 m = 3,927 cm.
4. Number of bulbs = 3,927 cm ÷ 25 cm = 157.08 bulbs -> 157 bulbs.
5. Number of packs = 157 ÷ 16 = 9.81 packs -> Round UP to 10 packs.

See detailed step-by-step working on Page 15 ->

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Step 1: Calculate diameter of the inner planting circle

75 cm = 0.75 m from the outer edge on BOTH sides of diameter:

Inner Diameter = 14 m − 0.75 m − 0.75 m = 12.5 m [M1]

(In centimetres: 1,400 cm − 75 cm − 75 cm = 1,250 cm)

Step 2: Calculate circumference of the planting circle

Circumference = π × diameter = π × 12.5 m

= 39.2699... m = 3,926.99... cm [M1]

(Or using π = 3.14: 3.14 × 12.5 = 39.25 m = 3,925 cm)

Step 3: Calculate the number of flower bulbs required

Bulbs are planted every 25 cm around circumference:

Number of bulbs = 3,926.99 cm ÷ 25 cm = 157.08 bulbs [M1]

Rounding to nearest whole plant = 157 flower bulbs

Step 4: Calculate number of packs needed (16 bulbs per pack)

Number of packs = 157 ÷ 16 = 9.8125 packs [M1]

Step 5: Round UP to the next whole pack

Packs can only be bought as whole boxes.

9 packs = 9 × 16 = 144 bulbs (not enough for 157 bulbs needed).

Therefore, Natalie must buy 10 full packs! [A1]

10

[5 Marks: M1 M1 M1 M1 A1]

(Total for Question 5 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6 Andrew is training to run in the New York Marathon.
The New York Marathon is 26.2 miles long.

The qualifying times to run in the New York Marathon are shown in the table below.

Age (years)	Time (hours : mins)
18 – 34	2 : 53
35 – 39	2 : 55
40 – 44	2 : 58
45 – 49	3 : 05
50 – 54	3 : 14
55 – 59	3 : 23
60 – 64	3 : 34
65 – 69	3 : 45
70 – 74	4 : 10
75 – 79	4 : 30
80 +	4 : 55

Andrew is 51 years old.
His current average running speed is 8 miles per hour.

Andrew thinks he will need to run at a faster speed to be able to qualify for the New York Marathon.

Is Andrew correct?
Show why you think this.

(3)

Step 1: Identify Andrew's qualifying standard from table

Andrew is 51 -> category "50 – 54" -> Qualifying Time = 3 hrs 14 mins [M1]

In total minutes: (3 × 60) + 14 = 194 minutes (or 3 + 14/60 = 3.233 hrs)

Step 2: Calculate Andrew's time at his current speed (8 mph)

Time = Distance ÷ Speed = 26.2 miles ÷ 8 mph = 3.275 hours [M1]

Convert decimal hours: 0.275 × 60 = 16.5 minutes

Andrew's current time = 3 hours 16.5 minutes (196.5 minutes)

Step 3: Compare times & reach conclusion

196.5 mins > 194 mins (Andrew takes 2.5 mins too long!)

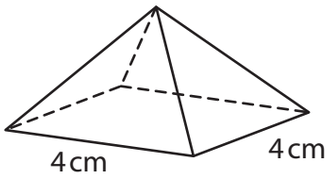
Yes, Andrew is correct — he must run at a faster speed to qualify! [A1]
Yes (takes 3h 16.5m vs 3h 14m)

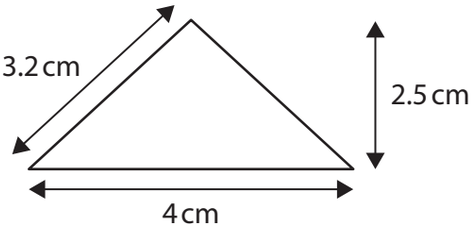
[3 Marks: M1 M1 A1]

(Total for Question 6 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

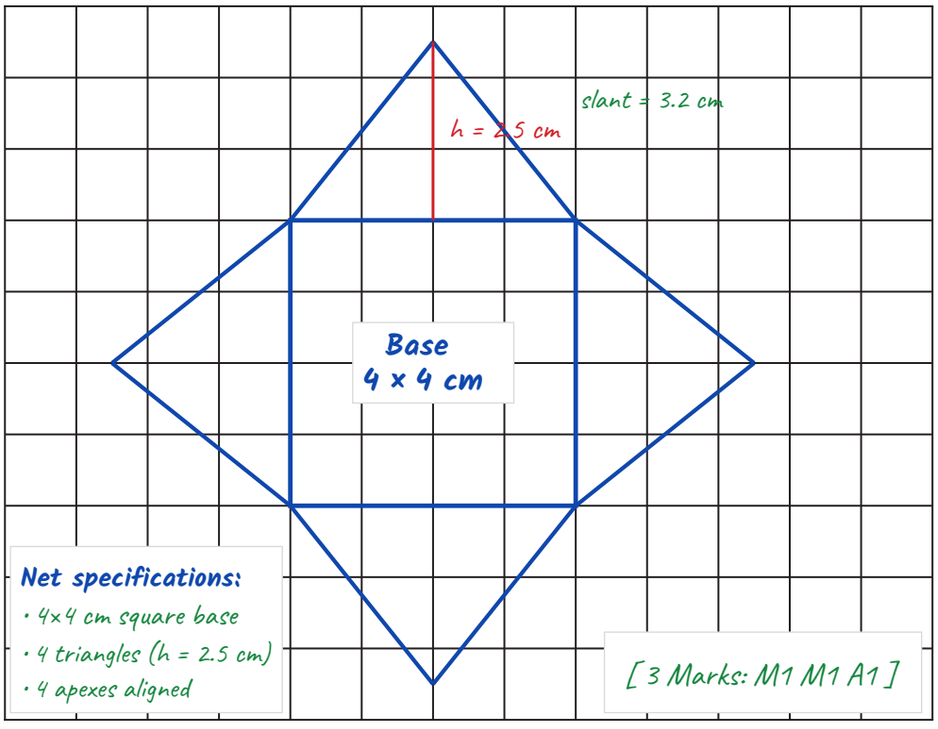
7 Here is a sketch of a square based pyramid and one of the triangular faces.





(a) Draw an accurate net of this pyramid on the centimetre grid below.

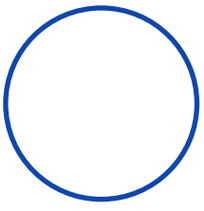
(3)



The base of a tin of paint is placed on the floor.
The tin of paint is in the shape of a cylinder.

(b) Draw a sketch of the plan view of the tin of paint.

(1)



Plan View = Circle

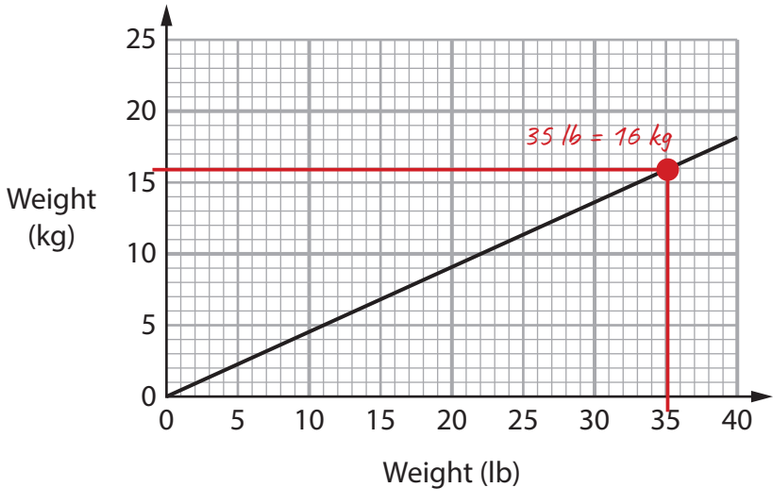
(looking directly from above) [1 Mark]

(Total for Question 7 is 4 marks)

8 Dave needs to make 3500 lbs of concrete mix using cement, sand and gravel.
He will mix cement, sand and gravel in the ratio 1 : 2 : 5 by weight.

Cement, sand and gravel are sold in kilograms.

Dave uses this graph to convert between lb and kg.



(a) Work out the weight, in kilograms, of cement, of sand and of gravel that Dave needs to buy.
You **must** show your working.

(5)

Step 1: Convert 3,500 lbs to kilograms using the graph

From graph: 35 lb = 16 kg [M1]

Scale factor = 3,500 ÷ 35 = 100

Total concrete mix in kg = 16 kg × 100 = 1,600 kg [M1]

Step 2: Divide into ratio parts: Cement : Sand : Gravel = 1 : 2 : 5

Total ratio parts = 1 + 2 + 5 = 8 parts [M1]

1 part = 1,600 kg ÷ 8 = 200 kg per part [M1]

See Page 19 for final component weights and ratio check ->

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Step 3: Calculate weight of each material Dave needs to buy

- Cement (1 part) = 1 × 200 kg = 200 kg [A1]
- Sand (2 parts) = 2 × 200 kg = 400 kg [A1]
- Gravel (5 parts) = 5 × 200 kg = 1,000 kg [A1]

Check: 200 + 400 + 1,000 = 1,600 kg (correct)

Cement	200	kg
Sand	400	kg
Gravel	1000	kg

(b) Show a check of your ratio calculation.

(1)

Reverse check using gravel:

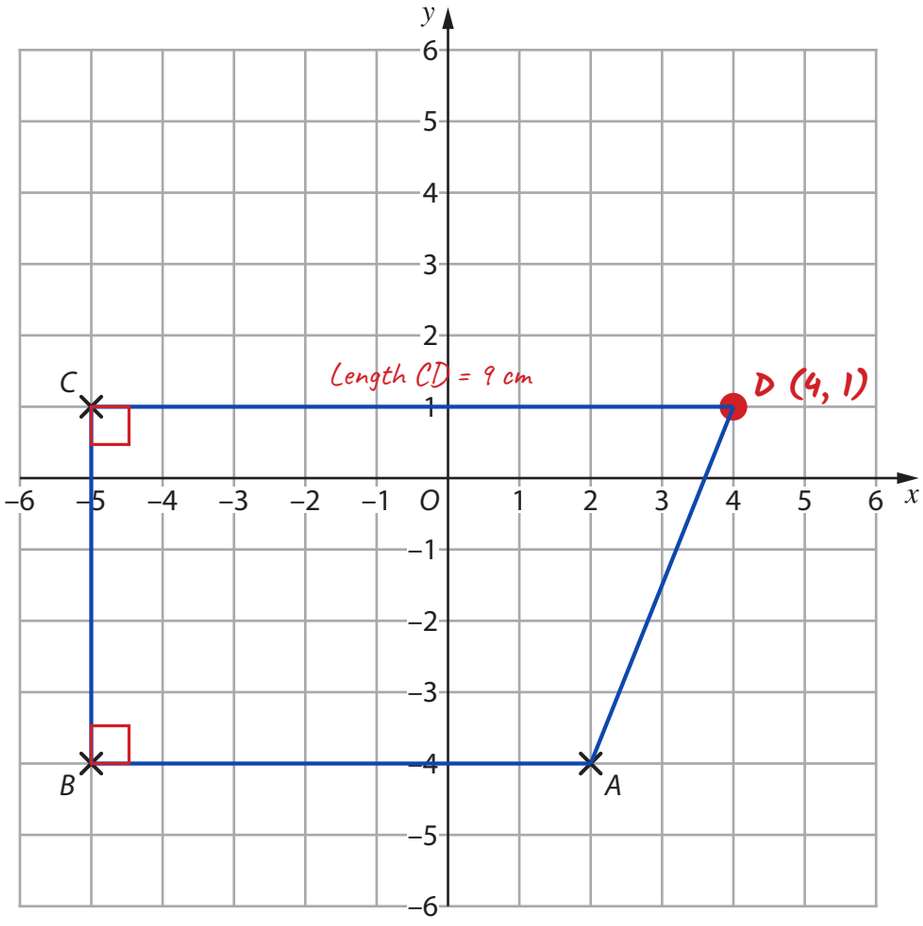
1,000 ÷ 5 = 200 kg (matches 1 part for cement)

Or sum check: 200 + 400 + 1000 = 1,600 kg (correct) [A1]

[6 Marks Total]

(Total for Question 8 is 6 marks)

9 Here is a centimetre grid.



(a) Write down the coordinates of point *B*.

(1)

(−5 , −4)

D is a point on the grid.

The trapezium *ABCD* has

- exactly 2 right angles
- one side of length 9 cm.

Step 1: Side BC is vertical (x = −5), so horizontal sides from B and C form right angles at B and C.

Step 2: Plot D at y = 1 with length CD = 9 cm

xD = −5 + 9 = 4 -> Point D = (4, 1) [2 Marks]

(b) Plot a possible position for point *D* on the grid.

(2)

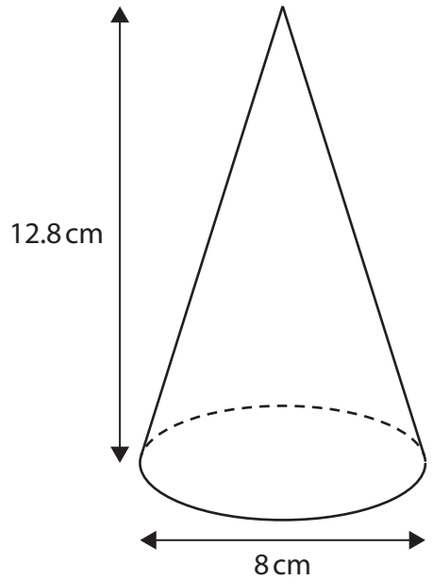
Point D plotted on grid at (4, 1) [A2]

[3 Marks Total: (a) 1, (b) 2]

(Total for Question 9 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10 Here is a cone.



$$V = \frac{1}{3}\pi r^2 h$$

V is the volume
r is the radius of the base of the cone
h is the vertical height

Work out the volume of this cone.
Remember to state the units.

(3)

Step 1: Find the radius of the circular base

Diameter = 8 cm

Radius r = Diameter ÷ 2 = 8 ÷ 2 = 4 cm [M1]

Step 2: Substitute into cone volume formula: V = (1/3) π r² h

V = (1/3) × π × (4)² × 12.8 [M1]

= (1/3) × π × 16 × 12.8

= (1/3) × π × 204.8

= 214.466... cm³ (or with π = 3.14: 214.36 cm³) [A1]

Tutor Note: Always remember to state units! Volume is in cm³.

214.4 cm³

[3 Marks: M1 M1 A1]

(Total for Question 10 is 3 marks)

11 Darius restores old furniture.
He has 12 chairs and 3 tables to restore.

Darius writes a list of the costs to restore all of the chairs and all the tables.

Costs	
tables and chairs	free
7 tins of wood oil	£12.50 each
10.5 metres fabric	£6.90 per metre
foam cushions	£82.50

Darius sells each table for £127 and each chair for £32

Work out the percentage profit Darius makes after selling all the chairs and all the tables.
Give your answer correct to 1 decimal place.

(5)

Summary of Costs & Selling Prices (Turn over for working):

- Costs: 7 × £12.50 + 10.5 × £6.90 + £82.50 = £242.45
- Sales: 3 tables × £127 + 12 chairs × £32 = £765.00
- Profit = £765.00 – £242.45 = £522.55
- Percentage Profit = (£522.55 ÷ £242.45) × 100 = 215.5%

See full step-by-step working on Page 23 ->

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Step 1: Calculate total restoration costs

- Wood oil: 7 × £12.50 = £87.50 [M1]
- Fabric: 10.5 × £6.90 = £72.45 [M1]
- Foam cushions: = £82.50

Total Costs = £87.50 + £72.45 + £82.50 = £242.45 [M1]

Step 2: Calculate total revenue from selling all furniture

- 3 tables @ £127 = 3 × £127 = £381.00 [M1]
- 12 chairs @ £32 = 12 × £32 = £384.00 [M1]

Total Revenue = £381.00 + £384.00 = £765.00 [M1]

Step 3: Calculate actual profit

Profit = Total Revenue − Total Costs

= £765.00 − £242.45 = £522.55 [M1]

Step 4: Calculate percentage profit

Percentage Profit = (Profit ÷ Total Costs) × 100

= (£522.55 ÷ £242.45) × 100 [M1]

= 2.155289... × 100 = 215.5289...%

Step 5: Round correct to 1 decimal place

The second decimal digit is 2 (< 5), so round down:

215.5289...% ≈ 215.5% [A1]

215.5 %

[5 Marks: M1 M1 M1 M1 A1]

(Total for Question 11 is 5 marks)

12 Jasmine is doing a survey about learning to drive.

The table shows information about the number of driving lessons 60 people had before passing their driving test.

Number of driving lessons	Frequency	Midpoint (x)	fx
1 to 15	6	8	48
16 to 30	18	23	414
31 to 45	25	38	950
46 to 60	11	53	583
		Total fx = 1,995	

Jasmine estimates that the mean number of driving lessons these people had before passing their driving test is more than 35 lessons.

(a) Is this estimate of the mean correct?
Show why you think this.

(3)

Step 1: Calculate midpoints and fx for each grouped class interval

- 1 to 15: Midpoint = (1 + 15) ÷ 2 = 8 : 8 × 6 = 48
- 16 to 30: Midpoint = (16 + 30) ÷ 2 = 23 : 23 × 18 = 414
- 31 to 45: Midpoint = (31 + 45) ÷ 2 = 38 : 38 × 25 = 950
- 46 to 60: Midpoint = (46 + 60) ÷ 2 = 53 : 53 × 11 = 583 [M1]

Sum of fx = 48 + 414 + 950 + 583 = 1,995 [M1]

Step 2: Calculate estimated mean

Estimated Mean = Total fx ÷ Total Frequency

= 1,995 ÷ 60 = 33.25 lessons [M1]

Step 3: Compare with Jasmine's estimate (> 35 lessons)

33.25 < 35 (or 60 × 35 = 2,100; 1,995 < 2,100)

Conclusion: No, Jasmine's estimate is incorrect. [A1]

No (33.25 < 35)

[3 Marks: M1 M1 A1]

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(b) Use a reverse calculation to show a check of your answer.

(1)

Tutor Check: Use reverse operation to check the estimated mean

Original calculation: 1,995 ÷ 60 = 33.25

Reverse Check: 33.25 × 60 = 1,995 (correct) [A1]

(Mental check: 33.25 × 6 = 199.5 -> 199.5 × 10 = 1,995)

One of these 60 people is chosen at random.

(c) What is the probability that this person had more than 30 lessons before passing their driving test?
Give your answer as a decimal.

(2)

Step 1: Find number of people who had more than 30 lessons

Classes with more than 30 lessons:

- **31 to 45 lessons: 25 people**
- **46 to 60 lessons: 11 people**

Total favourable people = 25 + 11 = 36 people [M1]

Step 2: Express probability as a decimal (Total = 60)

Probability = 36 ÷ 60 = 6 ÷ 10 = 0.6 [M1 A1]

0.6

[3 Marks: (b) 1 Mark, (c) 2 Marks]

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

P 6 8 4 7 4 A 0 1 7 2 0

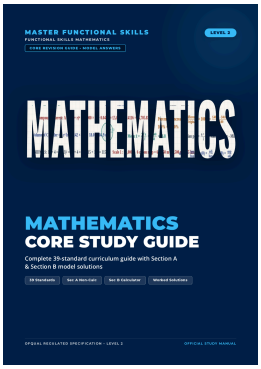
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Level 2 Exam



Core Curriculum Manual

Maths Level 2
Core Study Guide

- Full 39-standard curriculum
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

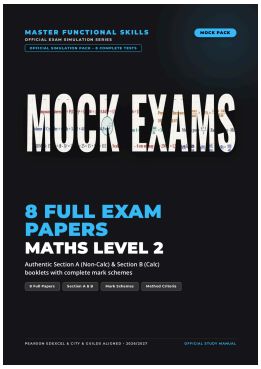


1,000+ Practice Drills

Maths Level 2
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF

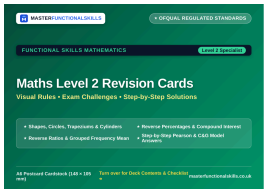


10 Realistic Past Papers

Maths Level 2
Official Mock Packs

- 10 full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 2
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk